

Independent Reproduction and Convergence Analysis of the Connes–Consani–Moscovici Zeta Spectral Triple

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Abstract

We independently implement the Connes–Consani–Moscovici (CCM) operator construction [1] in Rust at arbitrary precision (MPFR/GMP) and extend the headline result from 55 to **1019.0 measured matching decimal digits** on the first Riemann zeta zero at an HP-1000 requested-precision target with a 64-bit internal guard reserve, limited by the resulting guarded working precision. We characterize the construction’s convergence empirically: accuracy exhibits two-regime behavior in basis size N (linear growth, then saturation at the Weil eigenvalue ceiling ε_N); ε_N decays rapidly over the measured prime-count range (hundreds of decimal orders per doubling at fixed $N/\sqrt{\lambda^2} \approx 28$); and, across the sampled cutoffs, accuracy grows monotonically with λ and closely tracks ε_N . The smallest eigenvector’s even-symmetry (CCM’s Step 1 hypothesis) holds at every above-floor configuration tested up to $\lambda^2=1200$, HP-2000; the direct even-sector restriction is empirically unnecessary at these points — the unrestricted natural solve produces an even eigenvector and eigenvalues agreeing with the reduced-sector solve to hundreds of decimal digits. Remarkably, a 21×21 matrix from 6 primes already yields 21.585 matching digits. All results are reproducible from the accompanying source-available implementation. This work has been shared with the CCM authors and reviewed by the lead author, Professor Alain Connes.

1 Introduction

The Riemann Hypothesis (RH), asserting that all nontrivial zeros of the Riemann zeta function $\zeta(s)$ satisfy $\Re(s) = 1/2$, remains one of the central open problems in mathematics. Among approaches toward its resolution, the Hilbert–Pólya conjecture — positing the existence of a self-adjoint operator whose spectrum encodes the imaginary parts of the nontrivial zeros — has motivated extensive work in spectral theory and noncommutative geometry.

In November 2025, Connes, Consani, and Moscovici [1] proposed the finite-dimensional operator family $D_{\log}^{(\lambda, N)}$. Its spectra converge with striking numerical accuracy to the nontrivial zeros of $\zeta(1/2 + is)$. Their construction uses the Weil quadratic form built from the explicit formula, restricted to primes $p \leq \lambda^2$, and produces a $(2N+1) \times (2N+1)$ self-adjoint matrix whose smallest eigenvalue ε_N approaches zero as $\lambda, N \rightarrow \infty$. They reported accuracy of 2.5×10^{-55} on the first zero at ($\lambda^2=13, N=120$) with 200-digit working precision.

In this paper we:

1. Independently reproduce the CCM headline result on a Rust + MPFR/GMP software stack;
2. Extend the construction to ($\lambda^2=1000, N=800$) at an HP-1000 requested-precision target with 64 guard bits, achieving **1019.0 measured matching digits** on the first Riemann

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zero, limited by guarded working precision rather than by the construction’s intrinsic accuracy;

3. Systematically characterize the convergence behavior as a function of both N (basis size) and λ (prime cutoff);
4. Identify structural properties of the construction not discussed in [1], including empirical confirmation that the smallest eigenvector’s even-symmetry (Step 1 hypothesis) holds at every configuration tested above the precision floor; we explain the mechanism by which insufficient working precision produces erroneous mixed-symmetry artifacts when ε_N falls below the precision floor.

All computations are reproducible from the source-available implementation at <https://github.com/TeamXcelerator/ccm-reproduction-and-convergence>.

2 Mathematical Setup

We briefly recall the CCM construction, following the notation of [1].

2.1 The Weil Quadratic Form

Let $\lambda > 1$ and $L = 2 \ln \lambda$. The Hilbert space is $\mathcal{H}_\lambda = L^2([\lambda^{-1}, \lambda], du/u)$ with orthonormal basis

$$V_n(u) = \frac{1}{\sqrt{L}} e^{2\pi i n \ln(\lambda u)/L}, \quad n \in \{-N, \dots, N\}.$$

The Weil quadratic form QW_λ is a real symmetric $(2N+1) \times (2N+1)$ matrix with entries

$$\tau_{n,m} = W_{0,2}(V_n, V_m) - W_R(V_n, V_m) - \sum_{\substack{k=p^j \\ k \leq \lambda^2}} \frac{\log p}{\sqrt{k}} q(V_n, V_m)(\log k),$$

where $W_{0,2}$ is a closed-form rank-two pole correction, W_R is the archimedean (gamma-factor) contribution computed via Gauss–Legendre quadrature, and the prime-power sum encodes the von Mangoldt function $\Lambda(k) = \log p$ for $k = p^j$.

2.2 From Eigenvector to Eigenvalues

Let ε_N denote the smallest eigenvalue of τ and $\xi = (\xi_{-N}, \dots, \xi_N)$ the corresponding eigenvector, symmetrized to be even ($\xi_{-j} = \xi_j$) and normalized so that $\sum_j \xi_j = \sqrt{L}$. The positive eigenvalues of $D_{\log}^{(\lambda, N)}$ are $\nu_k = (2\pi/L)\sqrt{t_k}$, where t_k are the positive zeros of the rational function

$$R(t) = \xi_0 + 2t \sum_{j=1}^N \frac{\xi_j}{t - j^2},$$

located in the intervals $(k^2, (k+1)^2)$ for $k = 1, \dots, N$. Under the CCM construction, the ν_k approximate the imaginary parts of the nontrivial zeros of $\zeta(1/2 + is)$.

3 Implementation

Our implementation uses the Xcelerator Toolkit,¹ a publicly available library of high-precision spectral methods for analytic number theory, pinned to a tagged release for reproducibility. The

¹<https://github.com/TeamXcelerator/xcelerator-toolkit>

toolkit provides the core numerical primitives (Gauss–Legendre quadrature with disk-cached HP nodes, LU-based inverse iteration, Newton refinement) and the CCM-specific construction (Weil-form matrix assembly, eigenvector extraction, rational-function root-finding). The paper-specific repository contains only the CLI harness and reproduction scripts; all mathematical code is shared via the toolkit.

The toolkit is built on:

- **Rust** (edition 2021) for the main codebase;
- **rug** (v1.30, wrapping MPFR/GMP) for arbitrary-precision floating-point arithmetic;
- **nalgebra** (v0.33.3) for the f64 fast-path eigendecomposition (used only by the optional `--f64-only` mode for smoke tests; the HP publication path uses the toolkit’s own HP symmetric eigendecomposition in `xc-numeric::eigen`, verified at HP-1000 against a PARI/GP 2000-digit reference);
- **rayon** (v1.12.0 in the locked dependency graph) for parallel matrix assembly, Householder reduction, LU factorization, and eigenvalue-per-seed refinement.

Key algorithmic choices:

1. **Parity-aware eigenstate solve:** Paper claims use the direct reduced even-sector matrix. The default automatic solver first reuses an exact compatible eigenstate when available; on a cache miss it applies a retained-factor shift-invert Krylov/Rayleigh–Ritz method, optionally initialized by a compatible lower- N cached state. A residual-verified LU inverse-iteration route remains the conservative fallback when the targeted Ritz value is not isolated unambiguously. The unrestricted natural and conditional-projection routes use the full-space inverse-iteration implementation. Parity policy and solver route are part of artifact identity, so numerically equivalent states from different operation sequences cannot be confused in cache reuse.
2. **Halley’s method for $R(z)$ zeros:** The rational function $R(z) = \xi_0 + 2z \sum_{j=1}^N \xi_j / (z - j^2)$ has analytically available first and second derivatives, enabling Halley’s method (cubic convergence) as the default root-finder, seeded in the paper-claim scripts from a toolkit-owned reference dataset whose zeros were independently computed to 2,500 decimal digits using rigorous Arb interval arithmetic [3]. Their leading 1,000 digits were cross-validated against Odlyzko’s tabulation. Independent finite-source discovery is also available for research runs. The point solver distinguishes four outcomes: **Converged** (the requested correction target was met), **Stagnated** (a finite approximation stopped improving), **Approximate** (the iteration limit was reached with a finite value), and **Failed** (no usable finite value). Computed workflows retain finite nonconverged values with diagnostics; they are not mislabeled as converged or certified.
3. **Gauss–Legendre quadrature with disk cache:** GL nodes and weights at a given quadrature order and bit precision are computed once and cached for reuse across runs.

3.1 Reproducibility Infrastructure

The toolkit manages typed artifacts for quadrature rules, the archimedean and prime components, τ matrices, parity reductions, factorizations, Weil states, secular sources, root refinements, and validation evidence. Each artifact has a content-bound manifest whose semantic identity includes the numerical parameters, parity policy, algorithm route, precision contract, and dependency identities that can affect its value. Loads replay family-specific structural checks; incompatible or obsolete artifacts are rejected rather than silently reused.

Artifacts are stored first in a per-user local cache. Depending on the selected cache policy, compatible objects may also be resolved from managed public or authorized private shards; an ordinary user needs no credentials and computes locally on a miss. Publication is a separate author-only operation. Consequently, reproducibility does not depend on a particular remote fixture being present: a cache hit accelerates the run, while a miss executes the same content-identified numerical route.

4 Results

4.1 Reproduction of the CCM Headline

At ($\lambda^2=13, N=120$, 200-digit precision), our first eigenvalue is

$$t_1 = 14.134725141734693790457251983562470270784257 \dots$$

(displayed truncated to 42 digits; the full HP-1000 result matches the reference to 55.764 decimal digits, see Section 4.3) with absolute error 2.4363×10^{-55} against the independently computed 2,500-digit reference. This matches the CCM paper’s reported 2.5×10^{-55} to within rounding. The result is unchanged at an HP-1000 requested-precision target, confirming that the accuracy is controlled by $\varepsilon_N \approx 3.48399 \times 10^{-59}$ (a property of the construction at $\lambda^2 = 13$), not by arithmetic precision.

4.2 Extension to 1019 Measured Digits

Table 1: Measured matching digits across configurations, all at an HP-1000 requested-precision target with a 64-bit internal guard reserve. At $\lambda^2=1000$ the reported agreement saturates guarded working precision; the construction’s intrinsic ceiling is not resolved by this run.

Configuration	First-zero digits	20th-zero digits	Requested precision
$\lambda^2=13, N=120$	55.764	25.339	HP-1000
$\lambda^2=100, N=500$	460.09	422.44	HP-1000
$\lambda^2=1000, N=800$	1019.0	1020.3	HP-1000

Here HP-1000 denotes the requested accuracy contract, not the raw MPFR allocation. The implementation converts 1000 decimal digits to 3322 requested bits and appends 64 guard bits, giving 3386 working bits, or approximately 1019.3 decimal digits. Root acceptance is tested against the 3322-bit requested target; the guard reserve protects that target from cancellation and accumulated rounding error. The matching-digits diagnostic, however, compares the final values at the full guarded precision, so a converged Halley iteration can expose agreement within the reserve: 1019.0 digits for the first zero in the latest run. The 20th-zero relative metric reaches 1020.3 because it is normalized by $|t_{20}| > 1$, which can place it slightly above the raw decimal-digit equivalent of the bit allocation. These guard-space measurements do not redefine HP-1000 as a 1019-digit requested calculation; they report the additional agreement preserved by the internal working precision.

At configurations where ε_N is smaller than the working-precision floor (i.e., the construction’s intrinsic accuracy exceeds what the arithmetic can resolve), raising the working precision reveals additional matching digits approximately one-for-one. In general, accuracy is controlled jointly by λ (via ε_N), N (basis completeness), and working precision; all three must be scaled together to sharpen beyond a given level.

Table 2: Matching digits on the first five zeros vs. N at $\lambda^2=13$. Values are identical at HP-200 and HP-1000 across the entire sweep (see commentary below); the column shown is from the HP-1000 run.

N	Zero 1	Zero 2	Zero 3	Zero 4	Zero 5
10	21.585	16.335	11.856	5.1626	3.4915
20	35.795	32.261	30.011	26.504	24.870
30	44.506	41.267	39.262	36.208	34.844
40	50.294	47.152	45.225	42.304	41.015
50	53.827	50.721	48.822	45.948	44.685
60	55.299	52.203	50.312	47.451	46.195
80	55.654	52.560	50.670	47.811	46.556
100	55.735	52.641	50.751	47.892	46.637
120	55.764	52.670	50.779	47.921	46.666

4.3 Convergence in N (Basis Size)

Two regimes are evident:

1. **Rapid convergence** ($N = 10$ to $N \approx 60$): matching digits grow approximately linearly in N , with the per-step rate $\Delta(\text{digits})/\Delta N$ decaying smoothly from ~ 1.4 at $N=10 \rightarrow 20$ down to ~ 0.15 at $N=50 \rightarrow 60$. A single rate of ~ 0.96 digits per unit N is a useful average over $N \in [10, 40]$ but does not fit the entire pre-saturation regime; the rate at any specific N depends on how close N is to the saturation point.
2. **Saturation** ($N > 60$): accuracy plateaus at ~ 55.764 digits, limited by $\varepsilon_N = 3.48399 \times 10^{-59}$.

As with all empirical convergence rates, the precise per-step rate is a property of the specific (λ, N) regime sampled. We measured at $\lambda^2 = 13$ and $N \in \{10, 20, \dots, 120\}$; at different λ the sweep would show the same qualitative two-regime structure but with different absolute rates and a different saturation ceiling.

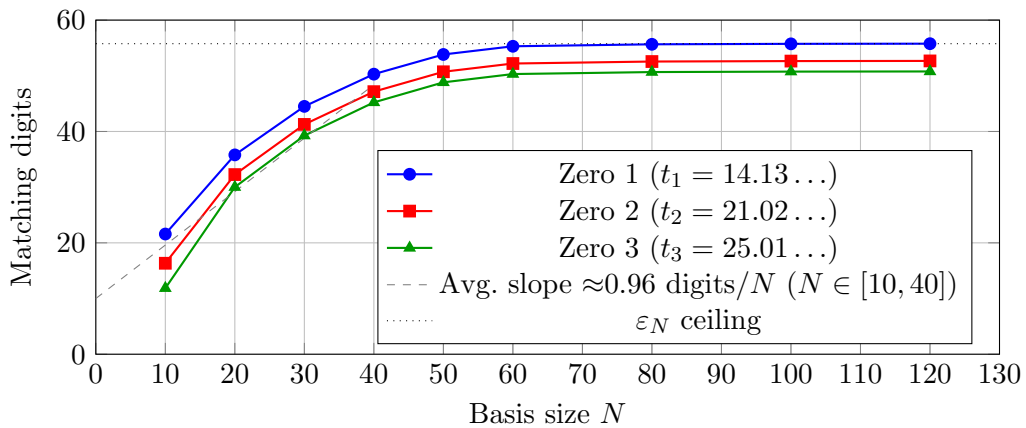


Figure 1: Convergence of matching digits vs. basis size N at $\lambda^2 = 13$, HP-1000. Linear regime transitions to saturation at the ε_N ceiling around $N \approx 60$.

Bonus observation (HP-200 vs HP-1000). At every N in this sweep, HP-200 and HP-1000 produce identical matching digits — confirming that the saturation at $N \gtrsim 60$ is intrinsic

to the construction (controlled by ε_N , not by working precision); see Sections 4.6 and 4.7 for the general mechanism.

4.4 The 21-Digit Result from a 21×21 Matrix

At $N = 10$ (matrix size 21×21), with only 6 primes ($\{2, 3, 5, 7, 11, 13\}$, corresponding to 9 prime powers entering the Weil form), the first Riemann zero is matched to **21.585 decimal digits**. This demonstrates notable numerical compression: more than 21 matching digits from a 21×21 matrix.

4.5 Rapid Empirical Decay of ε_N with Prime Count

Table 3: Smallest Weil eigenvalue ε_N across three configurations, each computed at working precision sufficient to resolve it (HP-1000 for $\lambda^2 = 13, 100$; HP-2000 for $\lambda^2 = 1000$), illustrating the rapid measured decay of ε_N with prime count.

λ^2	Primes $\leq \lambda^2$	ε_N	$-\log_{10} \varepsilon_N $
13	6	3.48399×10^{-59}	58.458
100	25	9.56481×10^{-464}	463.02
1000	168	$3.92215 \times 10^{-1264}$	1263.41

The measured decay is rapid. Quantified as decimal orders gained per doubling of prime count, the rate is large over the sampled range. Table 4 presents the above-floor series at a consistent basis ratio $N/\sqrt{\lambda^2} \approx 28$:

Table 4: Smallest Weil eigenvalue ε_N above the working-precision floor, with N scaled at the fixed ratio $N/\sqrt{\lambda^2} \approx 28$. HP-1500 for $\lambda^2=500-800$, HP-2000 for $\lambda^2=1000-1200$. The final column gives the decay rate in matching digits per doubling of prime count, computed against the preceding row; it ranges from approximately 437 to 613 over this sample and is illustrative rather than a measured law (see text).

λ^2	N	Primes $\leq \lambda^2$	ε_N	$-\log_{10} \varepsilon_N $	Digits/doubling
500	630	95	1.09117×10^{-929}	928.96	—
600	690	109	$2.22999 \times 10^{-1029}$	1028.65	502.7
700	740	125	$9.45898 \times 10^{-1116}$	1115.02	437.1
800	790	139	$1.35946 \times 10^{-1199}$	1198.87	547.5
1000	890	168	$1.52095 \times 10^{-1362}$	1361.82	596.1
1200	970	196	$6.79867 \times 10^{-1499}$	1498.17	613.1

These per-doubling values are illustrative, not a measured law. The rate depends on the parameterization (digits per doubling of prime count vs. per unit λ^2 vs. per prime added) and, critically, on how N is scaled with λ : ε_N is a function of both λ and N , so any rate quoted “per doubling of prime count” implicitly fixes a choice of $N(\lambda)$. We hold $N/\sqrt{\lambda^2} \approx 28$; a different scaling yields different numbers. What is directly supported by these measurements is that $|\varepsilon_N|$ decreases throughout the sampled family and gains hundreds of decimal orders over each listed prime-count doubling. Six finite samples cannot establish whether the asymptotic decay is polynomial, exponential, or super-exponential.

A rigorous derivation of the decay rate from the structure of the Weil quadratic form remains open. The empirical evidence shows a rapidly moving finite-configuration accuracy ceiling controlled by λ and N , but the precise functional form of $\varepsilon_N(\lambda, N)$ is not yet known.

4.6 Observed Monotone Convergence in λ at Fixed N

At fixed $N = 120$, accuracy on the first Riemann zero grows monotonically across the sampled λ cutoffs at every working precision tested, and closely tracks the smallest Weil eigenvalue $\varepsilon_N(\lambda)$. The construction’s behavior at fixed N is summarized empirically by the rule: matching digits $\approx -\log_{10} |\varepsilon_N(\lambda)|$ once the working precision is sufficient to resolve ε_N .

Table 5: First-zero accuracy vs. λ^2 at $N = 120$, comparing HP-200 and HP-1000 requested precision, each with a 64-bit internal guard reserve. First-zero accuracy agrees across the sweep. At $\lambda^2=100$, the HP-200 root retains 211.29 matching digits, although finer eigenvalue and sector diagnostics are close to the guarded working-precision floor. (Recall that ε_N depends on both λ and N ; the values shown here are at $N = 120$. At larger N for the same λ , ε_N continues to decay — e.g. at $\lambda^2=100, N=500$ the value is 9.56481×10^{-464} , see Table 3.)

λ^2	Primes $\leq \lambda^2$	ε_N at HP-1000	HP-200 digits	HP-1000 digits
13	6	3.48399×10^{-59}	55.764	55.764
20	8	2.50471×10^{-96}	92.817	92.817
30	10	2.19404×10^{-135}	131.80	131.80
50	15	6.89051×10^{-174}	170.19	170.19
100	25	3.48656×10^{-215}	211.29	211.29

For HP-200, the implementation converts the 200-decimal-digit request to 665 bits and adds 64 guard bits, yielding 729 working bits, or approximately 219.4 decimal digits. Root acceptance is still measured against the 665-bit requested target. The 211.29-digit result at $\lambda^2=100$ therefore uses part of the guard reserve without changing the nominal HP-200 accuracy contract. Diagnostics closer to the floor remain precision-sensitive. At $\lambda^2=100$, HP-200 gives $\varepsilon_N = 3.48676 \times 10^{-215}$, compared with 3.48656×10^{-215} at HP-1000. The corresponding complete-QR GapLog values are 4.246343 and 4.246284, agreeing to approximately five significant digits despite proximity to the HP-200 precision floor.

Two empirical regularities emerge:

1. **Observed monotone growth in λ .** At every sampled cutoff and working precision sufficient to resolve $\varepsilon_N(\lambda)$, accuracy increases as λ increases.
2. **ε_N -controlled accuracy.** Matching digits on the first Riemann zero match $-\log_{10} |\varepsilon_N|$ to within a constant offset (typically 2–3 digits) across the entire range. The Weil eigenvalue is not just an upper bound on accuracy — it is the operative quantity.

The relationship between accuracy and the prime cutoff is therefore mediated by ε_N : the cumulative change in the Weil form as prime powers enter the construction is reflected in the smallest eigenvalue, and the measured decrease in $|\varepsilon_N|$ translates into additional matching digits. This dependence is not perfectly featureless. Because a new term enters whenever the cutoff crosses a prime power, the finite matrix path has a localized slope kink at that threshold, precisely the event structure isolated by Groskin [4]. In sampled accuracy or eigenvalue curves these threshold effects can appear as small fine-scale perturbations on an otherwise smooth global envelope, rather than as the pronounced plateaus and jumps of a distinct staircase. Thus “smooth” here describes the overall trend; it does not mean differentiability or the absence of prime-power-scale structure. The CCM paper’s choice of ($\lambda^2=13, N=120$) at 200-digit requested precision yields 55.764 matching digits. With the same requested precision and the 64-bit guard reserve, increasing λ^2 to 100 raises the first-zero agreement to 211.29 digits, although sector diagnostics at that endpoint are already sensitive to the working-precision floor.

For practitioners, the implication is direct: choose λ to target the desired accuracy level via $-\log_{10} |\varepsilon_N|$, and choose precision $\gtrsim$ that level to avoid working-precision saturation. Within this

finite sweep we did not observe an additional λ -dependent failure mode once working precision and basis size were adequate.

4.7 Critical N at Fixed Working Precision

At fixed working precision, the construction reaches an apparent ceiling beyond which the computed eigenstate or root refinement can become floor-limited or stagnate. This ceiling is precision-dependent, not a property of the construction itself.

Table 6: Convergence at large N at HP-1000. Configurations that plateau at HP-200 continue to converge cleanly at HP-1000.

Configuration	ε_N at HP-1000	First-zero digits	5th-zero digits
$\lambda^2=50, N=200$	5.69795×10^{-221}	217.34	207.24
$\lambda^2=50, N=250$	4.51247×10^{-239}	235.45	225.42
$\lambda^2=100, N=300$	4.68074×10^{-368}	364.37	354.03
$\lambda^2=100, N=400$	6.12702×10^{-423}	419.28	409.04

The accuracy ceiling at fixed working precision is roughly (working precision)/ r , where r is the average per-step rate $\Delta(\text{digits})/\Delta N$ in the rapid-convergence regime. At $\lambda^2 = 13$ we measured $r \approx 0.96$ averaged over $N \in [10, 40]$ (Section 4.3); at other (λ, N) regimes r will differ, so the constant should be treated as a configuration-specific empirical value rather than a universal parameter of the construction. Once $r \cdot N$ exceeds the available digits, additional N yields no measurable improvement and inverse iteration begins to exhibit numerical artifacts. At 200-digit precision and $r \approx 0.96$ this gives a ceiling near $N \approx 210$; at 1000-digit precision the same configurations continue to gain digits well past $N = 400$ (the $\lambda^2=100, N=400$ row above resolves 419.28 first-zero digits).

This is the same precision-dependence phenomenon documented for λ in Section 4.6: the construction itself is well-behaved, but its accuracy at any finite N is bounded by the working precision used to compute it. Practitioners should choose precision $\gtrsim r \cdot N$ digits to avoid spurious saturation, with r measured (or estimated) for the specific (λ, N) regime of interest.

4.8 Even-Symmetry of the Smallest Eigenvector

The CCM proof’s Step 1 requires the smallest eigenvector ξ to be even (symmetric under $u \mapsto 1/u$). We measure the natural evenness $\|\xi - \gamma\xi\|/\|\xi\|$ of the unforced smallest eigenvector at working precisions sufficient to resolve ε_N above the floor:

Table 7: Symmetry properties of the natural (unforced) smallest eigenvector, each measured at working precision sufficient to resolve ε_N above the floor. At every configuration tested, the natural eigenvector is essentially even and the natural and even-sector eigenvalues agree to hundreds of decimal digits.

λ^2	N	Precision	Evenness deviation	Eigenvalue rel. diff.	Verdict
13	120	HP-1000	2.126×10^{-966}	1.107×10^{-962}	essentially even
100	500	HP-1000	6.665×10^{-563}	0 (at working precision)	essentially even
1000	800	HP-2000	7.634×10^{-763}	0 (at working precision)	essentially even
1000	890	HP-2000	1.563×10^{-664}	0 (at working precision)	essentially even
1200	970	HP-2000	2.948×10^{-528}	7.927×10^{-526}	essentially even

The $\lambda^2=100, N=500$ row was remeasured with the current 64-bit guard reserve; the earlier 16-guard-bit calculation gave 5.102×10^{-548} . The additional 48 working bits supply approxi-

mately $48 \log_{10} 2 = 14.45$ decimal digits of arithmetic headroom, consistent with the roughly 15-order reduction to 6.665×10^{-563} . This change lowers the residual symmetry noise without changing the smallest eigenvalue, its sign, or the “essentially even” conclusion.

The $\lambda^2=1000$ rows show the same guard-reserve effect. At $N=800$ the deviation falls from the earlier 2.941×10^{-748} to 7.634×10^{-763} , and at $N=890$ from 6.279×10^{-650} to 1.563×10^{-664} . Both improvements are approximately 14.6 decimal orders, consistent with the added 48 guard bits; the natural and even-sector eigenvalues remain numerically equivalent at the reported precision.

The $\lambda^2=1200$, $N=970$ row was likewise remeasured with the 64-bit guard reserve. Its evenness deviation falls from 2.927×10^{-514} to 2.948×10^{-528} , an improvement of approximately 14.0 decimal orders. The natural and even-sector eigenvalues are numerically equivalent at the attainable relative precision, agreeing to approximately 525 decimal digits with relative difference 7.927×10^{-526} . Their full 6708-bit representations need not be bit-identical: the natural full-space solve and the reduced even-sector solve follow different high-precision operation sequences, so roundoff at the working-precision floor need not produce identical terminal bit patterns. Moreover, 6708 bits provide approximately 2019 decimal digits, while $|\varepsilon_N| \approx 6.799 \times 10^{-1499}$ consumes roughly 1499 orders of magnitude; the resulting ~ 520 relative-digit headroom is consistent with the observed 525-digit agreement. Thus the nonzero difference reflects the arithmetic floor rather than a mathematical discrepancy between the two eigenvalues.

At every configuration tested above the precision floor — including $\lambda^2 = 1200$, far beyond the range previously reported — the natural smallest eigenvector is even to hundreds of digits of precision, and the natural and even-sector solves converge to numerically indistinguishable eigenvalues at the reported accuracy. In the current $\lambda^2=13$, $N=120$, HP-1000 reproduction, performed at 3386-bit working precision, their relative difference is 1.107×10^{-962} . CCM’s Step 1 hypothesis holds empirically at every tested configuration. We conjecture it holds universally: the smallest eigenvalue of QW_λ lies in the even sector for all (λ, N) .

The precision-floor artifact. When $|\log_{10} \varepsilon_N|$ approaches or exceeds the working precision in digits, the eigenvalue sinks below the arithmetic’s resolving power. In this regime the computed eigenvector is under-resolved and becomes sensitive to rounding within numerically indistinguishable directions. The result can *appear* non-even (deviation $\sim O(1)$), while the eigenvalues can *appear* to split between the natural and even-sector paths. Under the earlier guard policy, the HP-1000 $\lambda^2 = 1000$ configuration ($|\log_{10} \varepsilon_N| \approx 1264$, well past the available arithmetic floor) exhibits this artifact: deviation 1.87, eigenvalue split 24%. Increasing the requested precision to HP-2000 under the same policy at the *identical* ($\lambda^2=1000, N=800$) configuration collapses the apparent breakdown entirely: deviation 2.94×10^{-748} , with eigenvalues numerically equivalent at the reported precision. Thus the only variable in that controlled comparison is requested precision. The latest 64-guard-bit HP-2000 run uses 6708 working bits (approximately 2019.3 decimal digits, a floor ratio of approximately 1.60) and reduces the residual symmetry noise further to 7.634×10^{-763} , again with numerically equivalent natural and even-sector eigenvalues at the reported precision.

The even-sector restriction is empirically unnecessary. To further test whether the eigenvector is naturally even, we ran the full CCM pipeline (eigenstate solve $\rightarrow$ eigenvector $\rightarrow R(t)$ zero-finding $\rightarrow$ Riemann-zero matching digits) with the unrestricted natural full-space policy at configurations spanning $\lambda^2 = 13$ –1000, $N = 10$ –800, and HP-200 through HP-2000. At every tested above-floor point, the natural path produces eigenvalues and Riemann-zero approximations numerically indistinguishable from the reduced even-sector path at the reported accuracy. The direct restriction is therefore not needed to obtain the reported values at these points, although it remains the faster paper-reproduction default.

5 Discussion

5.1 Where the Construction’s Power Lies

The CCM construction’s measured accuracy is controlled jointly by three parameters:

- λ (via ε_N): determines the *ceiling* — how many digits are achievable;
- N (basis size): determines how quickly the ceiling is *reached*. At $\lambda^2 = 13$, the average gain over $N \in [10, 40]$ is approximately 0.96 digits per unit N , but the local rate decreases through this regime and is configuration-specific;
- working precision: determines whether the finite-configuration ceiling and its eigenstate can be resolved without floor artifacts.

Once N is large enough to resolve the eigenvector (empirically $N/\sqrt{\lambda^2} \gtrsim 25\text{--}30$ suffices at HP-1000 and above), further increases in N yield diminishing returns. The path to higher accuracy is primarily through larger λ (more primes), with N scaled proportionally to maintain adequate basis completeness.

5.2 The Prime-Weight Profile

Each prime power $k = p^j$ contributes weight $\Lambda(k)/\sqrt{k} = \log(p)/\sqrt{k}$ to the Weil form. For primes ($j = 1$), this is $\log(p)/\sqrt{p}$, whose derivative vanishes at $p = e^2 \approx 7.3891$ with maximum value $2/e \approx 0.73576$.

Table 8: Individual prime contributions $\log(p)/\sqrt{p}$ for the primes captured by $\lambda^2 = 13$. The maximum value $2/e$ is attained at the (non-prime) point $p = e^2 \approx 7.3891$; the integer prime 7 achieves 99.96% of this maximum.

p	$\log(p)/\sqrt{p}$	% of $2/e$
2	0.49013	66.62%
3	0.63428	86.21%
5	0.71976	97.83%
7	0.73548	99.96%
11	0.72299	98.26%
13	0.71139	96.69%
max = $2/e$ at $p = e^2$		100%

The CCM paper’s choice of $\lambda^2 = 13$ has the useful consequence that it captures the peak prime $p = 7$ to within 0.04% of the maximum value and includes both adjacent primes ($p = 5$ at 97.8% of the maximum and $p = 11$ at 98.3%) that lie within $\sim 2\%$ of the peak. The choice covers the entire near-peak cluster of primes.

Past the peak, $\log(p)/\sqrt{p}$ decays as a power law with a logarithmic factor (0.46052 at $p = 100$, 0.21844 at $p = 1000$, 0.092103 at $p = 10000$). Thus later primes still have visible individual weights at the finite cutoffs studied here. Their cumulative effect is consistent with the measured reduction in $|\varepsilon_N|$, but the scalar weight profile alone does not derive the eigenvalue-decay law; matrix interactions and the simultaneous scaling of N remain essential.

5.3 Implications for the Proof Strategy

Our findings have implications for the CCM proof strategy:

1. The even-symmetry of the smallest eigenvector (Table 7) holds at every configuration tested above the precision floor, up to $\lambda^2=1200$ at HP-2000; we conjecture it holds universally. The direct even-sector restriction is empirically unnecessary at the tested above-floor points — the natural eigenvector is even without it.
2. The saturation behavior (Table 2) shows that, within the sampled family, increasing N alone yields diminishing returns after the finite- λ ceiling is reached. Higher-accuracy tests therefore require increasing λ while keeping N adequately scaled.
3. The rapid measured decay of ε_N (Table 3) supplies empirical convergence evidence over the tested family, but both a rigorous proof and the asymptotic decay class remain open.

6 Conclusion

We have independently reproduced and significantly extended the CCM zeta spectral triple construction, achieving 1019.0 measured matching digits on the first Riemann zero at an HP-1000 requested target with a 64-bit guard reserve (limited by guarded working precision) and characterizing the construction’s convergence behavior in detail. The key findings:

- Two-regime convergence in N (linear growth then saturation at the ε_N ceiling).
- Rapid empirical decay of ε_N over the tested prime-count range.
- Observed monotone growth of accuracy across the sampled λ cutoffs, closely tracked by $\varepsilon_N(\lambda)$ at every tested working precision sufficient to resolve it.
- A precision-dependent accuracy ceiling at finite N , indistinguishable from the construction’s intrinsic ceiling unless the working precision is large enough to resolve ε_N .
- The smallest eigenvector’s even-symmetry (CCM’s Step 1 hypothesis) holds at every configuration tested above the precision floor, up to $\lambda^2=1200$ at HP-2000 — we conjecture it holds universally; the direct even-sector restriction is empirically unnecessary at these points, and apparent symmetry breakdown at large λ in floor-limited data is a precision artifact.
- The notable numerical compression at low N : a 21×21 matrix from 6 primes already produces 21.585 matching digits.

Together these provide a comprehensive empirical foundation for future theoretical work on this construction.

The observed compression (21 digits from 6 primes) suggests algebraic structure in the Weil quadratic form that is not yet theoretically understood. Explaining why this algebraic mechanism produces Riemann zeros with such precision remains an open problem whose resolution could advance the spectral approach to the Riemann Hypothesis.

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Data and Code Availability

The complete paper-specific source code and reproduction scripts are available at:

<https://github.com/TeamXcelerator/ccm-reproduction-and-convergence>

The shared mathematical library and content-bound reference-zero dataset are supplied by the Xcelerator Toolkit at:

<https://github.com/TeamXcelerator/xcelerator-toolkit>

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